

APPENDIX B: PYTHON CODE FOR RASPBERRY PI

```
import cv2

import torch

# Load lightweight
YOLOv5 Nano model
(auto-download)

model =
torch.hub.load('ultraly
tics/yolov5', 'yolov5n',
pretrained=True)

# Reduce model size
→ faster

model.conf = 0.4

model.iou = 0.4

# Start camera

cap =
cv2.VideoCapture(0)

# Low resolution for
speed

cap.set(3, 320)

cap.set(4, 240)

while True:

    ret, frame =
    cap.read()

    if not ret:

        break

# Inference

results =
model(frame,
size=320)

# Render results
```

```
annotated_frame =  
results.render()[0]  
cv2.imshow("Object  
Detection (Low Lag)",  
annotated_frame)  
if cv2.waitKey(1) &  
0xFF == ord('q'):  
break  
cap.release()  
cv2.destroyAllWindows()  
ws()
```